Response to the reviewer (fourth round)

A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

[Author name]

We thank the reviewer for the recommendation of minor revision and for the precise list of remaining corrections. No new computations were needed. All numerical results are unchanged from the third revision. The changes concern wording, scope statements and two table entries. Section and table numbers refer to the revised manuscript.

Comment 1 (status of the three-storey full-model estimate). *Preserve the distinction between the sampled-strata result and the residual-adjusted result, and avoid wording that implies a distribution-free bound for the complete full model.*

Response. We checked every place where the three-storey margin design is discussed (Abstract, Section 3.3, Table 6, Section 4.6 and Conclusion 5).

- Intervals and lower bounds that include the fitted t_3 residual are now called *model-conditional* confidence intervals and lower bounds wherever they appear. The same applies to the caption of Table 6.
- Section 4.6 previously said that the target is "met with 95% confidence under the t_3 residual model". It now says that the model-conditional lower bound (3.43) exceeds the target only under the assumed t_3 model, not under every tail model (t_2 gives 3.16), and that it is not a distribution-free bound.
- Conclusion 5 now reports the t_2 value next to the t_3 lower bound.

Comment 2 (code editions that define "code-compliant"). *Either update to the newer provisions or describe the study explicitly as a benchmark using AISC 360-16 and ASCE 7-05.*

Response. We retained the historical editions and now describe them explicitly as a fixed benchmark. We did not update to the newer editions, because the wind-load statistics of the probabilistic model are tied to the definition of the nominal wind load. From ASCE 7-10 onwards, wind loads are based on strength-level wind speeds with a load factor of 1.0, so a change of edition would change the nominal wind load and its statistics together, not just a load factor. The revision adds:

- a "Code basis" paragraph in Section 2.3. It states that AISC 360-16 and ASCE 7-05 are used as a fixed, internally consistent benchmark, that both have been superseded (AISC 360-22 and ASCE/SEI 7-22, now cited), and that the designs have not been checked against the current editions. It also defines "code-compliant" as compliance with the AISC 360-16 checks under the ASCE 7-05 combinations;
- the editions next to "code-compliant" in the Abstract, the list of formulations, the Discussion and the preamble of the Conclusions;

- in the code, a constant `CODE_BASIS` in `rbo/limit_states.py` that records the editions implemented.

Comment 3 (number of validated designs). *Table 1 still describes full second-order validation of eight designs; make the count consistent.*

Response. The "This study" row of Table 1 now reads "full P-Δ MC of 10 designs (8 principal + 2 three-storey variants)". We also found that the caption of Fig. 3 said "eight designs" although the figure shows ten. It now says ten designs and names the two three-storey variants, and the note in Section 3.3 refers to this figure as well.

Comment 4 (model factors in Table 3). *Table 3 labels the model factors "sensitivity only", although they are also used in RBDO-C-MU.*

Response. The model-factor row of Table 3 is generated by `make_figures.py`. It now reads "baseline: 0; sensitivity study: 0.05, 0.10; RBDO-C-MU (2 frames): 0.05". This states the c.o.v. for each use and agrees with Section 2.4 and Table 17.

Comment 5 (minimum-mass claims and the genetic algorithm). *Where global optimality has not been established, use wording such as best-found design.*

Response.

- Section 2.6 now defines "optimum", "optimal design" and "minimum-mass design" as the best feasible design found within the tested optimisation budget, not a proven global minimum. It notes that "minimum mass" in the title names the objective of the problem.
- In the Abstract, "a least-mass design at a stated reliability" became "the lightest design found, within the tested search budget, that meets a stated reliability".
- In the Discussion, "generally the least-mass way to meet both targets" became "among the designs compared, it generally gave the lowest mass that met both targets, with all masses being best-found designs within the tested optimisation budget, not proven global minima".
- Two further statements were reworded in the same way: Section 4.4 ("the best code-based design found is also the best reliability-based design found") and Conclusion 6.

Comment 6 (mass baseline in the Abstract).

Response. The Abstract now reads "2.7% more steel than the point-estimate design (5.2% more than the deterministic optimum)". Conclusion 5 states both premiums (+2.7% relative to RBDO-C and +5.2% relative to DDO).

Minor editorial items

- *Author information.* The placeholders remain in this version, because the author names and affiliations have not yet been supplied to us. They will be inserted before formal submission.
- *Cross-references.* The manuscript was recompiled after the changes to Tables 1 and 3. There are no undefined references or citations.
- *Tail-model terminology.* "Model-conditional confidence interval" is now used throughout (Comment 1).

- *Reproducibility.* The manuscript names the exact commit that produced it. The results are unchanged from the third revision. A release tag and DOI will be created on acceptance.
- *Cost interpretation.* Section 4.7 and Conclusion 6 now state that the cost index is an uncalibrated sensitivity model and has not been validated against actual project costs.
- *Confidence wording.* In the Abstract and Conclusion 3, "met both targets with 95% confidence" for the point-estimate designs now also names the baseline surrogate model.